

Data Analysis Report

Milestone 2: Advanced Data Analysis and Feature Engineering

Objective: Perform deeper data analysis and enhance feature selection and engineering to improve the model's predictive power.

1. Overview of Statistical Analysis

This part of the 2nd milestone focused on applying statistical tests to evaluate the relationship between customer behavior and churn risk scores. The analyses were conducted using a dataset that contains both engineered and original features representing user activity.

All hypothesis tests were conducted using a significance level (α) of **0.05**.

2. Oneway ANOVA Analysis

Purpose: Examine whether mean values of numeric features differ across different churn risk scores.

Assumption Check: Levene's Test for Homogeneity of Variance

Before applying ANOVA, Levene's test was used to assess if numeric features have equal variances across churn risk scores.

- Result:** Most numeric features failed the homogeneity assumption ($p\text{-value} \leq 0.05$), with the exception of:
 - age
 - days_since_last_login

ANOVA Test Application

Feature	p-value	Conclusion
age	> 0.05	Fail to reject H_0 – no significant difference
days_since_last_login	< 0.05	Reject H_0 – significant differences exist

Post-hoc Analysis (Bonferroni correction):

- **days_since_last_login:** Customers with churn scores 1 and 2 are statistically different from those with scores 3, 4, and 5.
- This supports a binary segmentation of churn risk:
 - Low risk (scores 1 & 2):
 - High risk (scores 3 to 5)

Visualization:

- **Bar Plot:** Mean `days_since_last_login` across churn risk scores
- **t-SNE Plot:** Visual clusters confirming the separation between high and low churn risk segments

3. Two-sample Welch’s t-Test (Unequal Variance)

Purpose: Evaluate if selected behavioral and financial features significantly differ between high and low churn risk customers.

Tested Features:

- `avg_time_spent`
- `avg_transaction_value`
- `avg_frequency_login_days`
- `points_in_wallet`

Segmentation: Customers were grouped into:

- **Low churn risk:** Scores 1 and 2
- **High churn risk:** Scores 3 to 5

Results:

Feature	p-value	Conclusion
<code>avg_time_spent</code>	< 0.05	Reject H ₀ – significant difference exists
<code>avg_transaction_value</code>	< 0.05	Reject H ₀ – significant difference exists
<code>avg_frequency_login_days</code>	< 0.05	Reject H ₀ – significant difference exists
<code>points_in_wallet</code>	< 0.05	Reject H ₀ – significant difference exists

Insight:

- All tested features significantly differ between churn groups, indicating they are strong predictors of churn behavior.

Visualization:

- **Boxplots:** Clear separation in distributions of all four features between high and low risk segments

4. Feature Insights and Recommendations

Feature	Key Insight
days_since_last_login	Strongest behavioral predictor of churn; supports binary risk segmentation
avg_time_spent	Lower average time spent indicates higher churn risk
avg_transaction_value	Financial engagement is a clear differentiator between churn groups
avg_frequency_login_days	Frequent login is strongly associated with high churn risk, implying these customers may be encountering service frustrations
points_in_wallet	High loyalty points correlate with retention

5. Dimensionality Reduction with t-SNE

- **t-SNE applied** to scaled numeric features revealed two distinct clusters aligned with churn risk levels.
- Confirms the statistical segmentation visually.

Conclusion

This milestone’s statistical analysis validated multiple strong relationships between engineered behavioral features and churn risk.